

Waste-polystyrene foams-derived magnetic carbon material for adsorption and redox supercapacitor applications

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ABSTRACT

The current impact of plastic waste on the environment and nature pushes for coming up with methods for their efficient reuse and recycle. Expanded polystyrene waste, massively produced worldwide, is presented in this work as a novel precursor of magnetic activated carbons (MAC) for the first time. A simple methodology based on the impregnation of EPS samples with Fe^{3+} in solution, followed by pyrolysis under Ar pressure and chemical activation is proposed. The as-prepared carbonaceous magnetic materials present nanometric phases of Fe^0 , Fe_3C , and Fe_3O_4 , and showed high specific surface area ($672 \text{ m}^2 \text{ g}^{-1}$) and total pore volume ($0.35 \text{ cm}^3 \text{ g}^{-1}$), one of the highest found in the carbonaceous magnetic materials literature. Their excellent textural, chemical and electrical properties, combined with the possibility of magnetically collection and regeneration after operation guaranteed an excellent performance of MAC in two different applications: as adsorbents of organic contaminants and as electrodes of redox supercapacitors.

1. Introduction

Activated carbons (AC) are a class of materials that find a wide range of applications due mainly to their textural properties. The high specific surface area and pore volume make these materials the most suitable for gas storage such as methane and hydrogen (Himeno et al., 2005), CO_2 capture (Pal et al., 2020), adsorption cooling (El-Sharkawy et al., 2015), heat pump applications (Pal et al., 2017), manufacture of supercapacitor electrodes (Lu et al., 2020), filters (Roegiers and Denys, 2020), medicines (Dasht Bozorg et al., 2020), sewage treatment (Wong et al., 2018), among others. Its importance is such that its market is growing and it is projected to reach USD 8.6 billion by 2026 (Industry Experts, 2020).

AC can be prepared by physical or chemical processes, from natural or synthetic precursors such as wood, coal, fruit cores and peels, phenolic resins, and plastic materials (Heidarinejad et al., 2020). Under

certain conditions, these materials can reach exceptional specific surface areas in the order of $3000 \text{ m}^2 \text{ g}^{-1}$ with a well-developed microporosity (de Paula et al., 2018). In recent years, a lot of research has been devoted to obtain ACs from plastic sources (Wang et al., 2020), with many works focusing on polyethylene terephthalate (PET) (Adibfar et al., 2014). In combination with the aim of obtaining carbonaceous materials with excellent textural properties, the reuse of plastic sources has an environmental interest in order to decrease pollution by obtaining new materials from waste.

Among the wide range of polymeric materials responsible for plastic pollution, polystyrene (PS) should be highlighted. PS is widely applied in the form of foam called expanded polystyrene (EPS), with a wide spectrum of applications ranging from the packaging sector to civil construction (Maharana et al., 2007). EPS is a versatile material, has excellent impact strength, acoustic isolation, easy processing, and,

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consequently, is massively produced worldwide. The problem is that the generation of EPS waste is proportional to its production, occupies a large volume due to its low density ($10\text{--}25\text{ kg/m}^3$), and increases disposal costs in landfills (Samper et al., 2010). In addition, PS is a non-biodegradable material and generates microplastics that have been found in rivers and oceans (Wang et al., 2019).

Considering the recycling and reuse of EPS, there are technical and logistical challenges (such as transportation costs and storage of large volumes of waste) that further contribute to the inappropriate disposal of the material (Maharana et al., 2007). Furthermore, the main recycling procedures are mechanical or chemical and have several drawbacks. For example, mechanical recycling has high energy consumption because it requires a combination of shear stresses and high temperatures (Maharana et al., 2007). Elevated temperatures still affect the structure of the polymer, impacting the appearance and mechanical strength of the recycled material (Maharana et al., 2007). Chemical recycling is an interesting alternative because some processes allow obtaining styrene monomers (Maharana et al., 2007). However, many chemical routes employ metallic catalysts and a large volume of expensive solvents.

In an attempt to make the recycling of EPS waste viable, obtaining products with higher added value appears as an excellent solution. Many industries grind and process waste into small pellets that are incorporated into building blocks or household items (Ramli Sulong et al., 2019). Even so, new strategic materials can still be envisioned using EPS as a source. Recently, de Paula et al. (2018) reported a simple methodology for obtaining high-quality AC from EPS residues, with high values of specific surface area ($2700\text{ m}^2\text{ g}^{-1}$) and total pore volume ($1.2\text{ cm}^3\text{ g}^{-1}$). This was made possible by performing the pyrolysis under Ar pressure, which keeps volatile products in the system, allowing their aromatization and char formation. The resulting materials showed a high adsorption capacity (1042 mg g^{-1}), evaluated with methylene blue as adsorbate. In addition, these ACs have a conductivity that enables their use as supercapacitor (SC) electrodes with high performance. Ortega et al. (2018) employed these ACs in the construction of conventional and hybrid SCs containing redox electrolyte, reaching specific cell capacitances of 54.3 and 77.9 F g^{-1} , respectively.

An elegant alternative for the optimization of AC applications is the formation of magnetic composites. Magnetically separable sorption materials are very useful in the environmental field (Rocha et al., 2020). Magnetic particles facilitate the separation of suspended material from the treated water (Rai and Singh, 2018), accelerate the coagulation of sewage (Booker et al., 1991), and can be useful as catalytic supports (Soltani and Zabihi, 2020). Several studies have already been published to obtain composites mainly containing Fe_3O_4 nanoparticles, that have low cost, biocompatibility, and superior magnetization at room temperature (Rocha et al., 2020). In this context, magnetic composites from plastic waste are strategic (Rai and Singh, 2018), but still little studied. It is necessary to develop simple processes to obtain carbonaceous magnetic materials with excellent textural properties, with few steps and that employ inexpensive products with low consumption. Also, there is a need for research focusing on the regeneration of adsorbent materials. To date, to the best of our knowledge, no study has reported obtaining magnetic activated carbons (MAC) from EPS with excellent textural properties, high adsorption capacities, and high regeneration efficiency.

Considering the energy storage field, the modern hybrid SCs require high-performance electrode materials to increase capacitance, energy density and, consequently, expand their application niche (Forouzandeh et al., 2020). Hybrid SCs accumulate charge by two mechanisms (by faradaic reactions and electric double layer) and can be constructed using redox shuttles dissolved in the electrolyte in high concentration (Zhang et al., 2020). However, this type of system requires electrodes with properties superior to those of conventional ones such as high specific surface area, greater electrical conductivity, high capacity for electrosorption of redox additives, and the ability to catalyze the faradaic surface-charge transfer (Santos et al., 2020). In this context, the synthesis of MAC, which contains metallic species in their structure, can

optimize their performance as SC redox electrodes, increasing electrical conductivity and providing new sites for electrosorption and faradaic charge transfer.

In this work we report a two-step conventional process to obtain magnetic activated carbon (MAC) from EPS residues. This is the first report on obtaining magnetic materials with a high specific surface area obtained from this plastic waste as a precursor. MAC performance was evaluated in two different applications: as an adsorbent for organic contaminants and as redox supercapacitor electrodes. Here, the physicochemical properties of MAC were studied and the application as an adsorbent was evaluated in terms of adsorption capacity (for different adsorbates and at different pHs), thermodynamics, kinetics, and reuse capacity (via thermal regeneration). As electrodes, MAC was evaluated for its electrical resistivity, specific capacitance, coulombic and energy efficiencies, and cyclability.

2. Material and methods

2.1. Synthesis of the activated carbon and magnetic activated carbon from EPS

For the synthesis of both AC and MAC, waste EPS was used as precursor without any previous treatment. The EPS waste used comes from electronic product packaging boxes and their characteristics were analyzed by thermogravimetric measures and discussed in the Supplementary Material (see Fig. S1).

Initially, 15 g of EPS was dissolved in 100 mL ethyl acetate and then the volume of solvent was reduced by heating in a bath of boiling water. Ethyl acetate was used to reduce the polymer volume, removing the air bubbles from the expanded polymer. After dissolving the polymer and reducing the volume of the solution, the mixture is subjected to a pyrolysis step. For this, the resulting mixture was introduced into a vertical furnace and pyrolyzed at $530\text{ }^\circ\text{C}$ (heating rate of $10\text{ }^\circ\text{C min}^{-1}$), under Ar, and at a pressure of 13 bar for 5 h. At the end of this step, approximately 2 g of solid (PS-Char) are obtained. For obtaining the MAC, a similar procedure was used, but 3.2 g of iron(III) acetylacetonate ($\text{Fe}(\text{acac})_3$) were added in the ethyl acetate-EPS mixture before the introduction in the vertical furnace. In this case, about 2.5 g of solid (PS-Fe-Char) were obtained.

The second and last step is the activation of the material to generate porosity. PS-Char and PS-Fe-Char were subjected to chemical activation using anhydrous potassium hydroxide (KOH) as activation agent. The mixture containing 2 g of char and 6 g of KOH was homogenized in a knife mill for 1 min. Then, the grounded material was subjected to a temperature of $800\text{ }^\circ\text{C}$ for 1 h, in a horizontal furnace, under a flow of N_2 , and a heating rate of $10\text{ }^\circ\text{C min}^{-1}$. After this process, the products were neutralized (using 10% (v/v) HCl solution) and washed with distilled water until pH 7 using a Soxhlet apparatus.

2.2. Characterization of the carbon materials

For the characterization of the iron phases present in MAC, X-ray powder diffraction (XRD) patterns were collected on a Siemens-D5000 diffractometer, using a Cu tube, in the angular range of $10^\circ < 2\theta < 80^\circ$ at a scan rate of $1.00^\circ\text{ min}^{-1}$. Mössbauer spectra were carried out in a CMTE model MA250 spectrophotometer, with constant acceleration, $^{57}\text{Co/Rh}$ source, using $\alpha\text{-Fe}$ as reference. Measurements were made of samples pulverized without application of external magnetic field and at room temperature. The spectra were fitted using a numerical program (NORMOS). Raman spectra were recorded from 800 to 3500 cm^{-1} on a Renishaw inVia Qontor Confocal Raman Microscope (Renishaw Instruments, England) using a 532 nm argon ion laser. X-Ray photoelectron spectroscopy (XPS) measurements were performed using a SPECS system under a pressure of 10^{-6} Pa with a non-monochromatic X-ray source (Mg K α radiation) operate at 12.5 kV and 100 W. CASA XPS was used for data interpretation. Shirley-background subtractions were

employed depending on the background shape. Peaks were fitted using GL(30) line shapes; a combination of Gaussian (70%) and Lorentzian (30%) shapes. All XP spectra were charge-corrected by referencing the fitted contribution of the C–C graphitic-like carbon in the C 1s signal to 284.6 eV.

The morphology and chemical composition of the MAC was investigated by scanning electron microscopy (SEM) in a QUANTA-FEG 3D FEI instrument equipped with an energy dispersive X-ray detector. Transmission electron microscopy (TEM) images were obtained on a JEOL JEM 2100F transmission electron microscope, equipped with a field-emission-gun, FEG, operating at 200 kV.

The total Fe content in MAC was obtained by flame atomic absorption spectrometry (Perkin Elmer AAnalyst Model 100). For this, 100 mg of MAC was refluxed in a 10% (v/v) HCl solution, at 150 °C for 2 h. Then, the mixture is filtered, and 5-fold diluted for further analysis.

The thermal and textural properties of AC and MAC were studied using thermogravimetric (TGA) and N₂ adsorption/desorption measures, respectively. TGA was carried out from room temperature to 800 °C under synthetic air flow (25 mL min⁻¹) at a heating rate of 5 °C min⁻¹, using a TA Instruments TGA Q5000. The textural properties of the carbon materials were obtained from their N₂ isotherms at 77 K, which were carried out in an AUTOSORB-1 (Quantachrome Instruments). Brunauer-Emmett-Teller model (BET) was used to calculate the specific surface area (S_{BET}) and Quenched Solid Density Functional Theory (QSDFT) was used to evaluate the pore size distribution.

2.3. Batch adsorption experiments (equilibrium and kinetics)

The adsorption capacities of MAC were evaluated for three different adsorbates: phenol (Ph), methylene blue (MB), and indigo carmine (IC). Initially, the optimum pH for the adsorption of each adsorbate was determined by adding 20 mL of adsorbate solution (3.7 mmol L⁻¹) in 15 mg of MAC at pHs 3, 5, 7, 9, and 11. The mixtures were shaken and kept vertically in a thermostatically controlled bath in the absence of light, at 298.15 K, for 48 h to ensure the equilibrium state. The collected aliquots were then diluted properly to evaluate the absorbance in the UV/VIS spectrometer (Shimadzu UV2550).

For all pHs evaluated, Zeta potential (ZP) values were obtained through electrophoretic mobility (EM) measurements in 1 mg L⁻¹ MAC suspensions. The data was obtained from an Anton Paar Litesizer 500 particle analyzer, and ZP values were calculated using the Smoluchowski equation, as the average of 3 independent measurements obtained as the mean of 100–1000 counts.

Adsorption isotherms were obtained at optimal pH of each adsorbate, 24 h of contact time, and in triplicate by mixing solutions (20 mL) containing different concentrations of the adsorbates with 15.0 mg of MAC. The kinetic experiments were also conducted with the collection of aliquots during the first hour of contact of the solution (5 mL; 0.05 mmol L⁻¹ of adsorbate) with the adsorbent (0.6 mg). Solution samples were taken at predetermined time intervals and centrifuged immediately at 4000 rpm. The supernatants of the solution sample were analyzed by UV/VIS spectrophotometry and the amount adsorbed (*q*) was calculated using Equation (1):

$$q = \left(\frac{C_0 - C_t}{m} \right) V \quad (1)$$

where *C*₀ and *C*_{*t*} correspond to initial concentration and concentration at the sampling time *t* (mg L⁻¹), respectively, *m* is the mass of MAC (g) and *V* is the volume of the solution (L).

2.4. Thermal regeneration experiments

To assess the regeneration capacity of MAC, five consecutive adsorption/desorption cycles were performed using a procedure with the following steps: (i) At each adsorption step, 15 mg sample were

added to 20 mL of MB solution (3.5 mmol L⁻¹), and the suspension was stirred for 24 h at 25 °C. The mixture was centrifuged and the supernatant was analyzed by UV/VIS spectrophotometry in order to determine the amount of MB adsorbed. (ii) For the desorption steps, the solid was dried at 120 °C for 5 h and then was calcined at 400 °C (10 °C min⁻¹) under the N₂ flow rate (100 mL min⁻¹) for 2 h in a vertical tube furnace to remove the dye. Finally, for comparisons with MAC, the regeneration capacity of AC was determined using the same methodology.

2.5. Preparation of the electrodes and electrochemical experiments

The electrodes were prepared in a mortar by mixing 90 wt% of the respective activated carbon and 10 wt% of polytetrafluoroethylene. The formed pellets have a diameter of 1.13 cm², thickness of 0.2 mm, and weight of 10 mg. The electrical resistivity of the electrodes was measured using a four-point probe system (Keithley Instruments, Model 238 High Current Source Measure Unit).

The electrochemical experiments were carried out with complete cells prepared in a three-electrode configuration using two pellets containing the active materials as electrodes (one as a working electrode and the other as a counter electrode) and Ag/AgCl/KCl (3.5 M) as a reference electrode. Measurements were performed on “T-type” Swagelok cells employing an aqueous solution of 0.07 M MB/1 M H₂SO₄ as electrolyte. Cyclic voltammetry and cyclic chronopotentiometry tests were performed in a Biologic VMP3 Potentiostat/Galvanostat (USA) at 25 °C. From galvanostatic experiments, the specific electrode capacitance (*C*_{spec}), coulombic efficiency (*ε*), and energy efficiency (*η*) were calculated from Equations (2)–(4), respectively.

$$C_{\text{spec}} = \frac{2I \int V dt}{m V_{\text{discharge}}^2} \quad (2)$$

$$\epsilon = 100 \times \frac{\Delta t_{\text{discharge}}}{\Delta t_{\text{charge}}} \quad (3)$$

$$\eta = 100 \times \frac{\left(\int V dt \right)_{\text{discharge}}}{\left(\int V dt \right)_{\text{charge}}} \quad (4)$$

where *I* is the current applied, *t* is the time, *V* is the electrode potential, *m* is the mass of active material in the electrode, *Δt*_{discharge} is the discharge time, *Δt*_{charge} is the charge time and *V*_{discharge} is the maximum potential avoiding the ohmic drop.

3. Results and discussion

3.1. Characterization of the magnetic AC composite

The phase composition of the MAC was examined by X-ray diffraction (Fig. 1a). Three distinct crystalline phases within the MAC structure can be identified: Fe⁰, Fe₃C e Fe₃O₄. The signals located at 2θ = 44.7° and 65.0° are attributed to the planes (110) and (200), respectively, corresponding to Fe⁰ (JCPDS 87–0721). Diffraction peaks at 2θ = 18.3°, 30.1°, 35.4°, 37.1°, 43.1°, 53.6°, 57.1°, and 63.0° are indexed to (111), (220), (311), (222), (400), (422), (511), and (440), respectively, corresponding to the cubic phase of magnetite (Fe₃O₄) (JCPDS 01–1111). The Fe₃C phase was identified by the (020), (211), (102), (122), (230), and (301) planes (JCPDS 35–0772), which are related to the peaks at 2θ = 26.4°, 42.9°, 43.7°, 51.8°, 54.4°, and 58.0°, respectively. For all identified phases, these peaks appear broad, suggesting a small crystallite size. The average grain size, estimated by Scherrer's equation, was found in the order of nanometers (ca. 13.0 nm for Fe⁰, 15.4 nm for Fe₃O₄, and 10.5 nm for Fe₃C).

The Mössbauer spectrum for MAC is shown in Fig. 1b. The same species identified by XRD were confirmed by Mössbauer analyses. The spectrum is composed of three sextets that are attributed to Fe₃O₄ (in

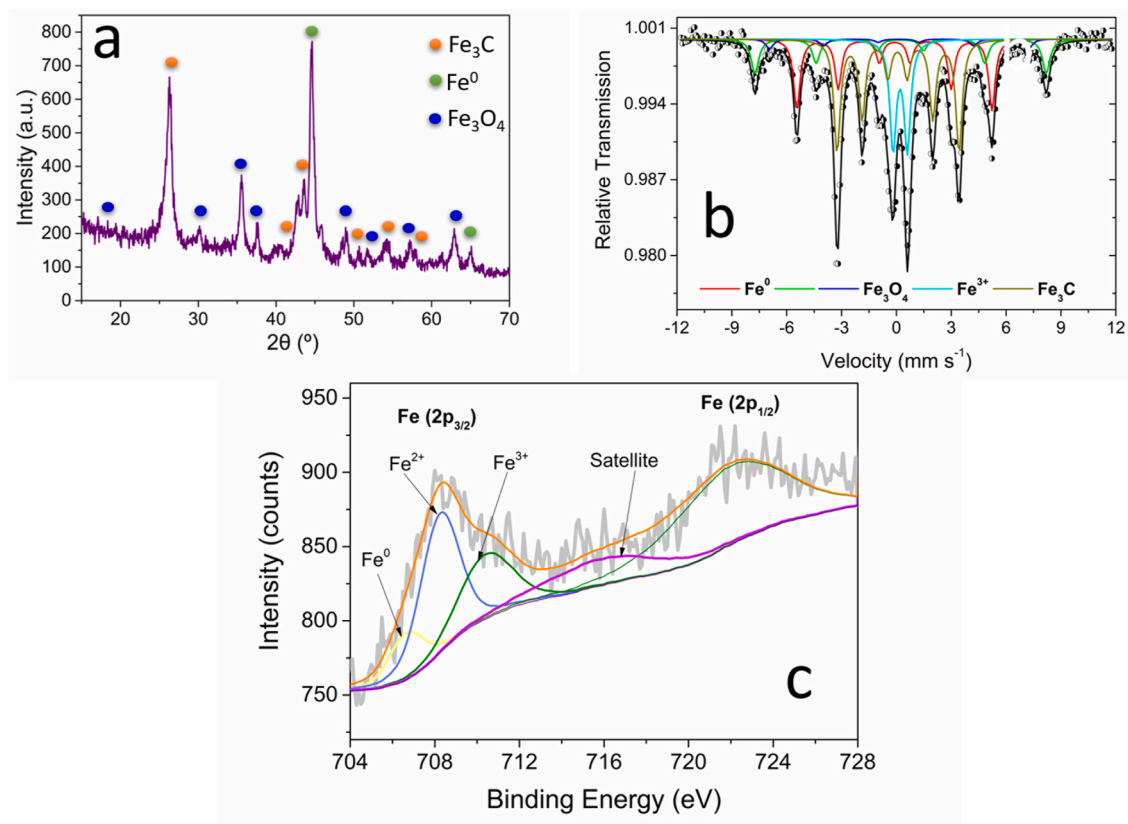


Fig. 1. (a) Powder XRD for MAC. (b) Mössbauer spectrum for MAC at 298 K. (c) X-ray photoemission spectra (Fe 2p) of MAC.

green and dark blue) and Fe^0 (in red). The doublet (in light blue) may be associated with the presence of the Fe^{3+} ions in a superparamagnetic oxide phase. The presence of Fe_3C is associated with the quartet marked in brown.

The hyperfine parameters obtained from the Mössbauer spectrum are summarized in Table 1. The relative areas of the signals correlate with the percentage of each phase present. The iron phases are distributed in MAC with the following composition: 24% Fe^0 , 20% Fe_3O_4 , 38% Fe_3C , and 18% of Fe^{3+} dispersed. It is important to note that the Fe^{3+} phase dispersed/nanostructured within the MAC structure was not observed in the obtained diffractogram. This has been observed in other works (Pinto et al., 2018, 2019). The Fe^{3+} superparamagnetic phase can be present in two forms: i) nanometric phases (<20 nm) and/or ii) amorphous phases of iron oxides/hydroxides (Murad, 2010).

MAC synthesized in this work has a higher Fe_3C content (38%). Molina-Sabio et al. (2013) reported this effect on the synthesis of activated carbons prepared by the pyrolysis of ethylene tar with dissolved metal acetylacetonates. According to the authors, the presence of iron in the activation stage induces carbon recrystallization and carbide formation.

Table 1
Hyperfine parameters obtained by Mössbauer spectroscopy for MAC.

Phases	δ (mm s ⁻¹) (± 0.05)	$\Delta/2\epsilon q$ (mm s ⁻¹) (± 0.05)	Bhf (T) (± 0.2)	AR (%) (± 1)
Fe^0	0.00	0.00	33.1	24
Fe_3O_4	[Fe^{3+}] < $\text{Fe}^{3+}/\text{Fe}^{2+}$ >	0.31 0.00	49.3 43.9	15 5
Fe_3C	Fe^{3+}	0.17	20.8	38
Fe^{3+}		0.30	0.78	18

δ = isomeric displacement in relation to $\alpha\text{-Fe}$, ϵ = quadrupolar displacement, Δ = quadrupolar unfolding, Bhf = hyperfine field, AR = relative areas of the subspectrum; [] octahedral and < > tetrahedral coordinated iron.

In previously published works, non-magnetic oxides (hematite and goethite) (Oliveira et al., 2002), iron hydroxides (Mohan et al., 2011), and maghemite (Oliveira et al., 2002) were obtained in the composites. Non-magnetic phases should be avoided or some procedures are required to convert them to magnetite in order to increase the magnetization of the material (Oliveira et al., 2002). Since MAC has 82% of magnetic iron phases (Fe , Fe_3O_4 , and Fe_3C), this attests that the synthesis conditions used in this study were favorable for obtaining high-quality magnetic composites.

XPS measurements were also used to characterize the iron species present in MAC, and corroborate the results found by XRD and Mössbauer spectroscopy. The XPS spectra (Fig. 1c) shows the peaks corresponding to $\text{Fe}(2p_{3/2})$, $\text{Fe}(2p_{1/2})$, and a satellite centered at 715.8 eV (characteristic of Fe^{2+} species) (Kuivila et al., 1988). The $\text{Fe}(2p_{3/2})$ peak is narrower and has a larger area than the $\text{Fe}(2p_{1/2})$ peak. This is a known feature and is due to the spin orbit (j-j) coupling. $\text{Fe}(2p_{3/2})$ and $\text{Fe}(2p_{1/2})$ have degeneracy of four and two states, respectively (Radu et al., 2017). The curve fitting to $\text{Fe}(2p_{3/2})$ shows three peaks associated with Fe^0 , Fe^{2+} , and Fe^{3+} . The Fe^{2+} and Fe^{3+} peaks are centered at 708.2 and 710.4 eV, respectively, and are associated with the magnetite phase. Considering the signals from the Fe_3C , it is known that its $\text{Fe}(2p_{3/2})$ binding energy is similar to that of Fe^0 (Kuivila et al., 1988).

Fig. 2 shows the SEM and TEM micrographs obtained of the MAC. SEM images show that MAC powders present irregular shape with a wide particle size distribution (ranging from sub-micrometer to micrometer size particles). The EDX mapping of individual elements showed that nanometric structures containing Fe are uniformly distributed throughout the whole material. This can be seen by observing the same pattern in the distribution of signals from C and Fe (Fig. 2b). The homogeneity obtained in the iron phases distribution is expected for the type of synthesis employed (Castro et al., 2010). This is one of the advantages of dissolving metallic compounds together with the carbon precursor (Castro et al., 2010).

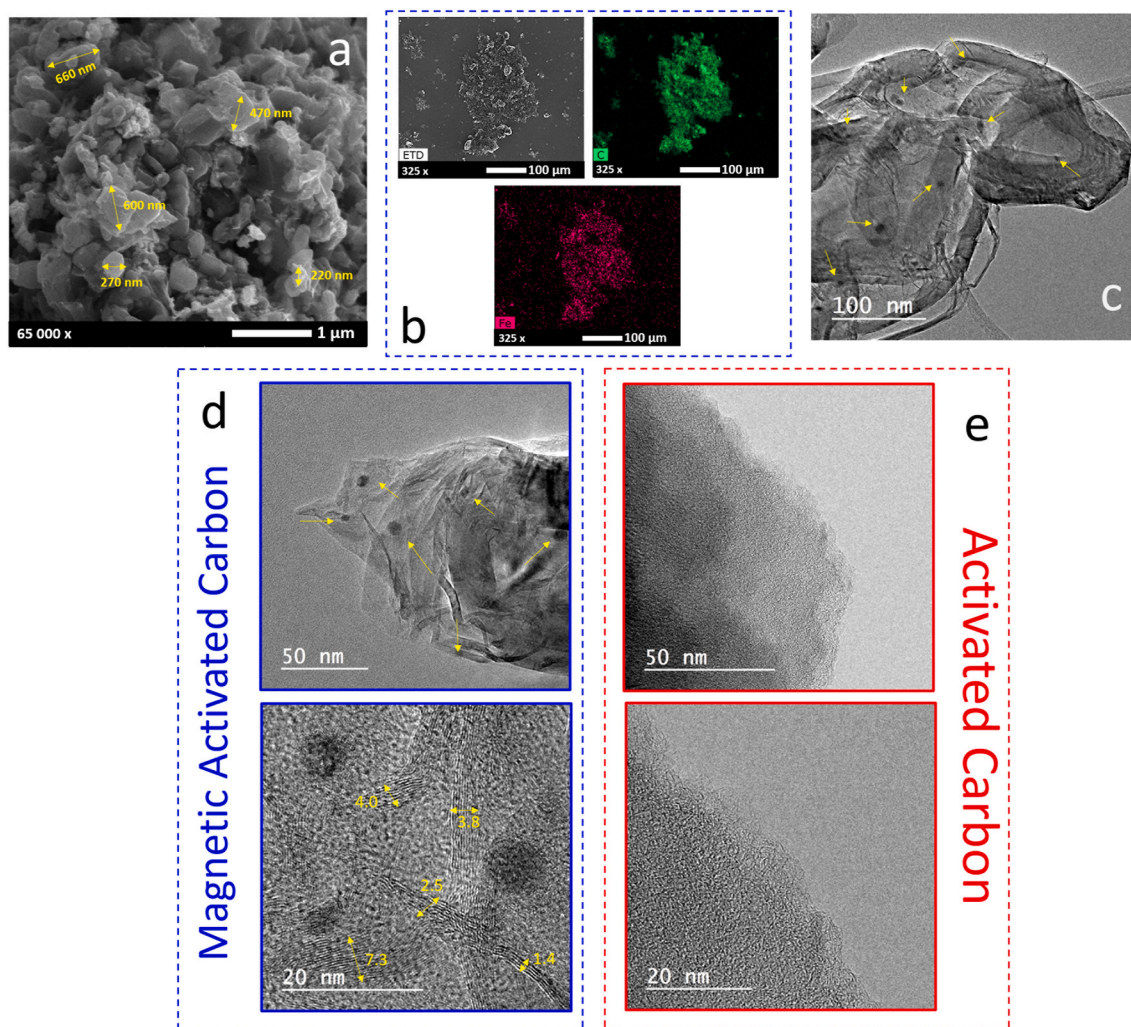


Fig. 2. (a) SEM image for MAC and (b) EDX mapping analysis for C and Fe. TEM images for (c, d) MAC and (e) AC.

In addition, the average size of the iron species, around 10 nm, estimated by Scherrer's equation, is confirmed by the TEM images. These iron species are embedded in the carbonaceous matrix and can be distinguished as the darkest points in the images (some of these points were marked in yellow in Fig. 2c and d). In the images of greater magnification, the comparison between MAC and AC is interesting. AC has a highly disordered structure of non-graphitized carbons with individual layers that are twisted, bent and randomly arranged (Fig. 2e). On the other hand, MAC images show several graphitic bundles in this microstructure, formed with layers organized in different thicknesses. Therefore, it can be inferred that the presence of iron during the synthesis of MAC induces the graphitization of the carbonaceous matrix. This effect has been observed in other studies (Molina-Sabio et al., 2013).

Quantified by flame atomic absorption spectroscopy (FAAS) the MAC from EPS waste contains up to 14.2 wt% of metallic species (Fe^0 , Fe_3O_4 , Fe_3C , and Fe^{3+}). As noted in the TEM images, the presence of these iron species impacts the carbon matrix structure, inducing graphitization. This can also be verified by analyzing the thermal stability of the composite and comparing it with AC produced in the absence of the metallic compound. Fig. 3 shows the curves (a) TGA and (b) dTGA for AC and MAC, respectively.

The first thermal event occurs between 25 and 120 °C and is associated with the physisorbed water removal (Amaral et al., 2020). The residues left after 1000 °C are very different for MAC and AC. AC leaves 6% of residual mass due to the absence of iron species. For MAC, heating

in the presence of oxygen causes the oxidation of iron species, gaining mass and remaining in ashes (32.1% of residual mass). More interesting is the main thermal decomposition that is related to the degradation of the carbon phase. In MAC, this degradation starts at higher temperatures (~329 °C) than AC (~264 °C). This increase in thermal stability is significant and indicates that MAC has a more organized structure and, consequently, more stable.

Raman spectroscopy was also employed to assess structural differences in the carbon phases of MAC and AC (Fig. 3c). For both materials, the D-band (around 1345 cm^{-1}) and G-band around (1600 cm^{-1}) are present. The first is related to defects in the carbon structure whereas the latter is related to more organized structures (Ferrari and Robertson, 2000). The position and width of these bands depend on the degree of carbonization and structure disorders (porosity, crystallite size, etc.). In this way, the structural differences between carbon materials can be detected by the D/G intensity ratio (I_D/I_G). In addition, second-order bands can be observed at higher wavenumbers ($2600 - 3000 \text{ cm}^{-1}$) attributed to overtones (2D) and a combination of known lattice vibration modes (G + D) (Sadezky et al., 2005). Before considering the differences between the AC and MAC spectra, it is important to note that the Raman spectra were obtained at different points in each sample. For AC, all these points have similar Raman scattering and the average of the spectra has I_D/I_G equal to 0.93. On the other hand, MAC has regions with structural organization similar to that of AC ($I_D/I_G = 0.92$), but it also has domains with greater organization ($I_D/I_G = 0.53$). Therefore, these results corroborate with the observation of the graphitic regions by TEM

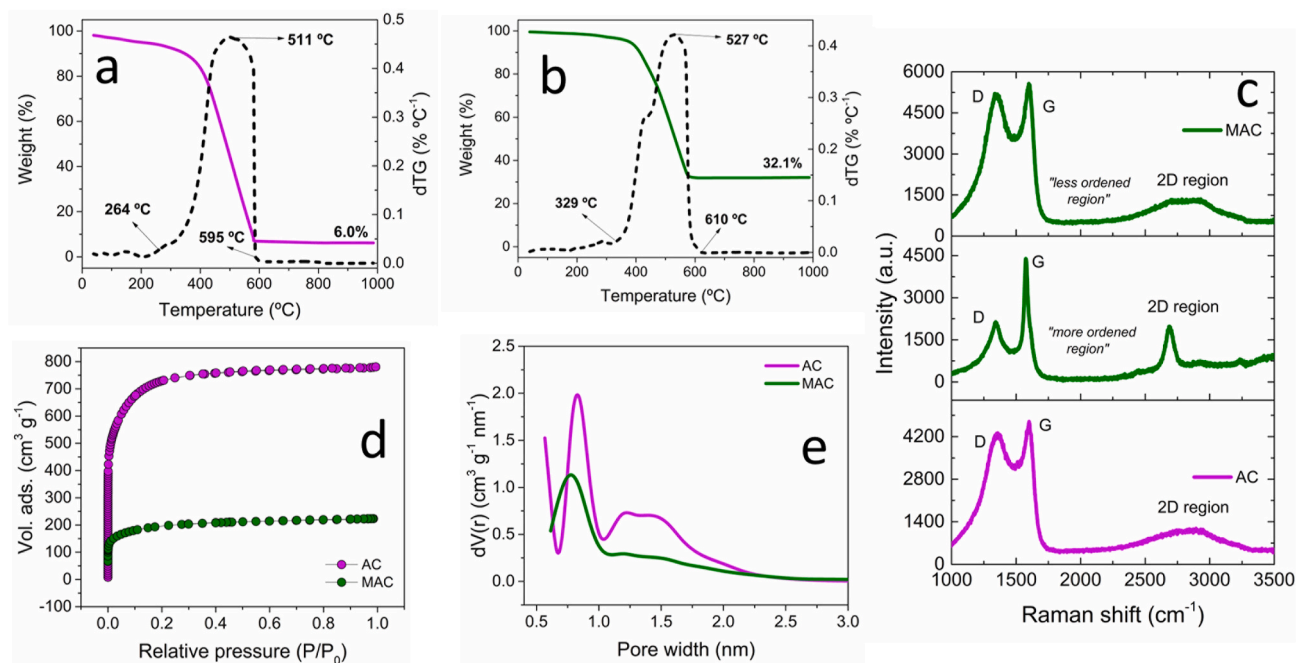


Fig. 3. TGA and dTGA curves for (a) AC and (b) MAC. (c) Raman spectra, (d) N₂ adsorption-desorption isotherm and (e) Pore size distribution by applying Quenched Solid Density Functional Theory (QSDFT) for the carbon materials.

and justify the greater thermal stability of MAC observed in the TGA measurements.

Among all the physicochemical properties of porous materials, textural properties are one of the most important. AC are the most commercially used materials in contaminant removal processes and supercapacitor electrodes. What connects these quite different applications is the need for high specific surface areas and high pore volumes. Fig. 3 shows the (d) N₂ adsorption/desorption isotherms and the (e) pore size distribution (PSD) for AC and MAC. The textural properties for AC from EPS are previously known (de Paula et al., 2018), but it is important to compare with MAC to know the effects of adding metallic compounds on the development of porosity in the magnetic material. The profile of the isotherms is similar for both carbons, being typical of microporous materials (de Paula et al., 2018). The isotherms showed a wide knee in the transition from low to high relative pressures that indicates a wide distribution of pores. This is corroborated by the PSD calculated by QSDFT (Fig. 3d). QSDFT model is considered more suitable for the analysis of the adsorption isotherms of carbon materials having various degrees of pore wall roughness and surface defects, in relation to other commonly used ones such as NLDTF (Kwiatkowski et al., 2019). PSD is also similar for both materials where its pores present diameters less than 2.5 nm. Also, both materials have a high content of small micropores with a distribution centered at approximately 0.8 nm.

Despite the similarities in the profile of the isotherm and the PSD, it is clear that AC exhibits a much higher N₂ adsorption capacity in relation to MAC. However, this is expected since in the MAC the iron species do not contribute to N₂ adsorption and have a high mass fraction (Oliveira et al., 2002). Thermogravimetric measurements also indicate that the carbon phase in MAC has structural differences in relation to AC, and this can also affect the development of porosity. In this context, quantitative textural parameters were obtained from the N₂ isotherms and summarized in Table 2.

In agreement with the isotherms, AC has the highest S_{BET} and V_{T} values (2562 m² g⁻¹; 1.21 cm³ g⁻¹). The same textural parameters are significantly reduced in MAC, with S_{BET} and V_{T} values corresponding to 672 m² g⁻¹ and 0.35 cm³ g⁻¹, respectively. The ratios between $S_{\text{micro}}/S_{\text{T}}$ and $V_{\text{micro}}/V_{\text{T}}$ are similar for the two carbons and indicate that about

Table 2

Quantitative Textural Data for CA and CAM obtained from N₂ adsorption/desorption isotherms at 77 K.

Material	S_{BET} (m ² g ⁻¹)	S_{micro} (m ² g ⁻¹)	V_{T} (cm ³ g ⁻¹)	V_{micro} (cm ³ g ⁻¹)	V_{meso} (cm ³ g ⁻¹)
AC	2562	1531	1.21	0.93	0.28
MAC	672	426	0.35	0.27	0.08

S_{BET} is the specific surface area; S_{micro} is the specific surface area of the micropores; V_{T} is the total pore volume; V_{micro} and V_{meso} are the micropores and mesoporous volume, respectively.

60% of the specific surface area and approximately 80% of the pore volume are related to microporosity. Despite the inevitable reduction in S_{BET} and V_{T} for MAC, these values are high and comparable to the excellent carbonaceous adsorbents reported in the literature (see Table S1 - supplementary material). The comparison is more suitable with magnetic AC composites from different sources including plastics such as PET (highlighted in Table S1).

3.2. Adsorption of organic contaminants on MAC

The adsorption capacities of MAC for Ph, MB, and IC were evaluated. Fig. 4a shows the maximum adsorption capacity (q_{max}) for each adsorbate at different pHs. Note that each adsorbate has an optimal pH where the amount adsorbed has the highest value (hatched bars). The highest q_{max} values are found at pH 3, 7, and 9 for IC (623.2 mg g⁻¹), Ph (115.7 mg g⁻¹), and MB (281.0 mg g⁻¹), respectively. Although the adsorption of these compounds can occur by different mechanisms, electrostatic interactions are the major ones affected by pH. Except at pH = 3 ($Z_{\text{P}} = +6.67$ mV), MAC has a negative surface charge at the other studied pHs ($Z_{\text{P}} < -20.0$ mV) (Fig. 4b). This justifies the higher adsorbed amount of the MB cationic dye at alkaline medium (pH > 7), while the anionic dye IC is preferably adsorbed at the most acidic medium evaluated. For Ph, the pH changes do not significantly affect q_{max} (ranging between 100 and 117 mg g⁻¹), and pH = 7 was considered the optimal pH for further studies.

It is important to highlight that the q_{max} values for the evaluated

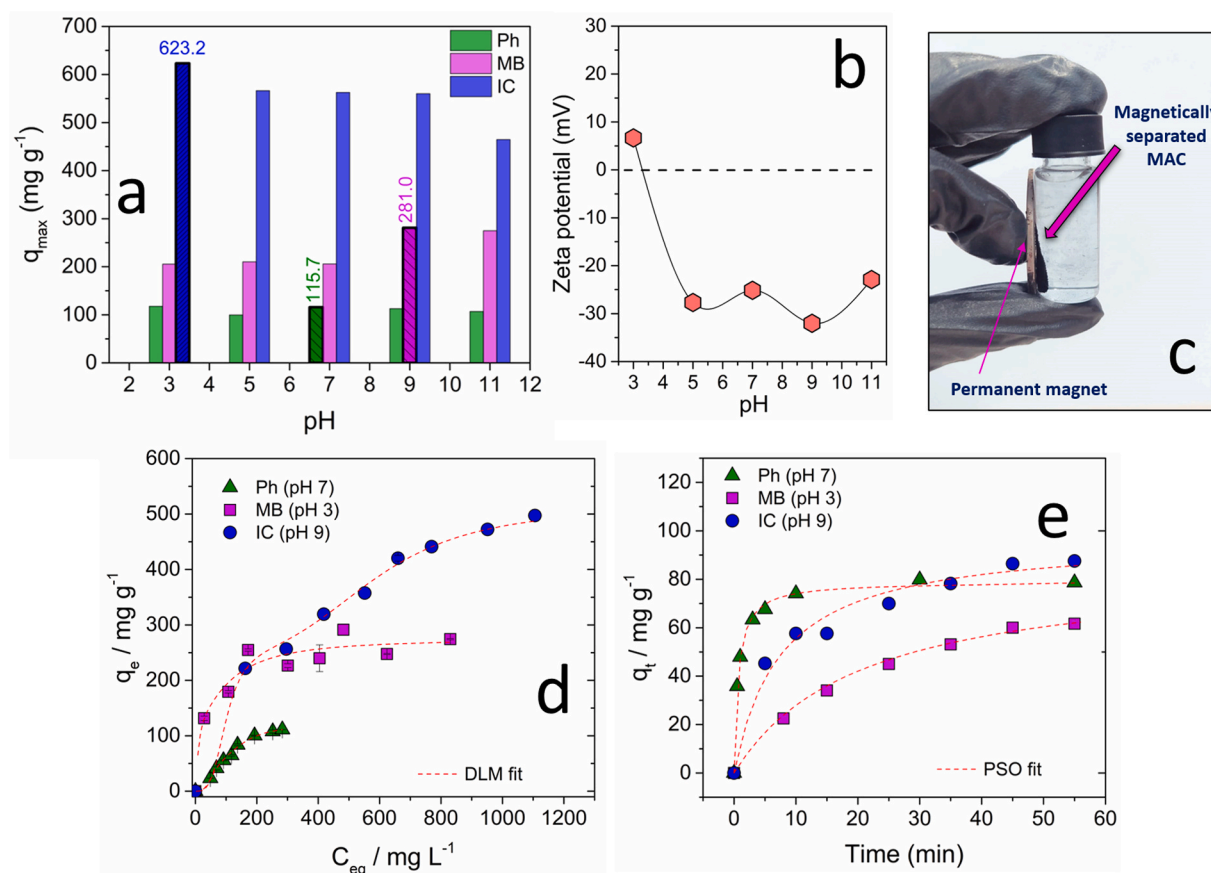


Fig. 4. (a) Evaluation of pH on the maximum amount of adsorbed MB, IC, and Ph on MAC at 298.15 K. (b) Zeta potential of MAC samples at different pHs. (c) Magnetic separation of MAC in solution using a permanent magnet. (d) Adsorption isotherms and (e) kinetic analysis for the adsorption of MB, IC, and Ph on MAC at 298.15 K.

adsorbates on MAC are among the highest reported in the literature for magnetic adsorbents (see Table S2 - supplementary material). Furthermore, the iron species present in MAC are not leached into the solution and, consequently, the material does not lose its magnetic property (Fig. 4c). The concentration of iron species in the solution was measured by FAAS, after the adsorption experiments at pH 3. Even in the most acidic medium studied (pH = 3) – a condition in which the formation of $Fe^{2+}(aq)$ is thermodynamically favored – the presence of iron species was not detected, suggesting that they are encapsulated in the pore structure of MAC or has great affinity for the carbon interface.

Adsorption isotherms were obtained at the optimal pH for each adsorbate and are shown in Fig. 4d. The analysis of the isotherms was performed with statistical mechanics models based on the grand canonical partition function (Khalfoui et al., 2002). With this approach, the isotherms were fitted using the single-layer model (SLM) and the double-layer model (DLM) (Mohamed et al., 2018). A description of the formalism and the parameters obtained in the adjustment are available in the Supplementary Material. Considering the Langmuir and Freundlich models, widely used in the literature to fit isotherms data, they have

several limitations since the assumptions of these models are simplistic, restrict, and do not allow the calculation of thermodynamic properties with a single temperature (Ortega et al., 2017).

The experimental isotherms were better adjusted using the DLM model. Then, the theoretical saturation capacity (q_{max}^{DLM}), the number of molecules per adsorption site (n), and thermodynamic parameters (ΔH , $T\Delta S$, and ΔG) up to the saturation limit were calculated and summarized in Table 3.

The adjusted q_{max}^{DLM} values are in agreement with the q_{max} values obtained experimentally at the optimal pH (Fig. 4a), with a greater deviation for IC. The adsorption process of all adsorbates studied on MAC is spontaneous ($\Delta G < 0$), exothermic, and entropically driven. In solid-liquid interface, the configurational entropy of water molecules that are desorbed and released into the bulk solution has a greater contribution (Ortega et al., 2017). The parameter $n > 1$ indicates that more than one molecule can be adsorbed per interaction site (Mohamed et al., 2018). This result suggests that a large fraction of the adsorbate molecules, especially IC and Ph, are oriented perpendicular to the adsorption site on the MAC's surface. The ΔH values suggest that the intensity of the

Table 3

Thermodynamics properties of adsorption calculated by DLM from the adsorption isotherms of MB, IC, and Ph on MAC. Parameters of pseudo-second-order kinetic model.

Adsorbate	Thermodynamic parameters at 298.15 K						Kinetic parameters of PSO at 298.15 K	
	q_{max}^{DLM} (mg g ⁻¹)	n	ΔH (kJ mol ⁻¹)	$T\Delta S$ (kJ mol ⁻¹)	ΔG (kJ mol ⁻¹)	r^2	k (g mg ⁻¹ min ⁻¹)	r^2
MB	274.85	1.60	-21.03	111.48	-132.51	0.95	5.75×10^{-4}	0.99
IC	512.38	3.49	-68.68	124.11	-192.79	0.99	1.38×10^{-3}	0.97
Ph	117.49	3.49	-21.27	29.03	-50.29	0.99	1.90×10^{-2}	0.99

IC-MAC interactions is stronger in relation to MB-MAC and Ph-MAC. It is interesting to note that for Ph and MB, the ΔH values are similar (-21 kJ mol^{-1}) but the T ΔS parameter is much smaller for the Ph adsorption ($29.03 \text{ kJ mol}^{-1}$).

Table 3 presents the rate constants (k) resulting from the adjustment of the pseudo-second-order model (PSO) applied to the kinetic data presented in Fig. 4e. It is important to point out that the PSO kinetic model showed the best fits for the adsorbates studied in this work (in relation to the pseudo-first-order model) and this is a common empirical behavior found in the literature for the removal of organic contaminants by carbonaceous adsorbents (Saleh and Al-Absi, 2017). The sorption kinetics experiments were also performed under conditions of optimum pH for each adsorbate and the results showed that the adsorption is favored for Ph, which has a k value ($1.90 \times 10^{-2} \text{ g mg}^{-1} \text{ min}^{-1}$) higher than the other adsorbates in more than one order of magnitude (5.75×10^{-4} and $1.38 \times 10^{-3} \text{ g mg}^{-1} \text{ min}^{-1}$ for MB and IC, respectively).

3.3. The regeneration and recycling test

Regeneration is one of the most important characteristics when considering the feasibility of an adsorbent material. There are different methodologies reported in the literature for the regeneration of carbonaceous adsorbents. Ideally, such procedures should decompose the adsorbate and preserve the structure and physicochemical properties of the adsorbents. For MB decomposition (used as an adsorbate in recycle tests), previous studies indicate a thermal treatment (400°C and in N_2 atmosphere) as appropriate (Niu et al., 2018). TGA measurements, carried out in the N_2 atmosphere, show negligible mass losses for both AC and MAC at 400°C (Fig. 5a).

Fig. 5b shows the retention of adsorption capacity of MB for AC and MAC after five adsorption/regeneration cycles. In the first cycle, MAC preserve high MB adsorption capacities (97%) in relation to its initial capacity (281.0 mg g^{-1} at $\text{pH} = 9$). After 5 runs, the adsorption capacity is reduced to 59%. For the non-magnetic AC, thermal regeneration dramatically reduces the MB adsorption capacity by 45% in the second cycle. At the end of 5 runs, the MB adsorption capacity is only 14% in relation to the original material (1065 mg g^{-1} at $\text{pH} = 9$). Thus, we can conclude that MAC obtained from EPS are adsorbents with good regeneration capacity.

3.4. MAC performance as a redox supercapacitor electrode

In this session, the performance of MAC as an electrode of redox-SCs has been evaluated. Initially, the electrical conductivities of AC and MAC were measured using the four-point method (Fig. 6a). High

electrical conductivities are essential for good performance as electrodes of electrochemical devices. The values found for AC and MAC correspond to 9.0×10^{-3} and $1.8 \times 10^{-2} \text{ S cm}^{-1}$, respectively. The greater conductivity of MAC in relation to AC is a consequence of the higher content of graphitic domains in its structure, as observed in the tests discussed above.

Cells containing MAC as electrodes were built and tested in an aqueous acidic electrolyte containing MB dissolved. To allow proper comparisons, cells containing the non-magnetic AC were also built under the same conditions. The latter, synthesized from EPS, has already been evaluated as redox-SC electrodes in previous works and presented excellent electrochemical performance (Ortega et al., 2018). Fig. 6 shows the CV curves for the cells containing (b) AC and (c) MAC as electrodes. Note that, for both electrodes and all scan rates evaluated, the signals from the MB redox reactions overlap the typical box-like curve of EDL capacitors. Moreover, the currents generated in voltammograms containing MAC were more intense and contained the signals related to the $\text{MB}^{2+}/\text{MB}^+$ couple (formal potential centered at $+0.15 \text{ V}$ vs. $\text{Ag}/\text{AgCl}/3.5 \text{ M KCl}$) and the signals from $\text{Fe}^{3+}/\text{Fe}^{2+}$ couple centered at $+0.45 \text{ V}$ (vs. $\text{Ag}/\text{AgCl}/3.5 \text{ M KCl}$). The higher currents generated by the transfer of electrons from MB indicate that MAC has a greater capacity for electrosorption of the redox additive. This may be a consequence of its greater electrical conductivity, which favors the transfer of electrons at the electrode/electrolyte interface.

Considering the galvanostatic measurements, the curves for electrodes of redox-SCs presented plateaus centered on the formal potential of the redox shuttles (highlighted in Fig. 6d and e). The curves for MAC showed two plateaus relative to the $\text{MB}^{2+}/\text{MB}^+$ and $\text{Fe}^{3+}/\text{Fe}^{2+}$ redox couples, in agreement with the voltammetric measurements. These faradic processes increase the electrode charge accumulation capacity, which is expressed in its specific capacitance. Fig. 6e shows the specific capacitances (C_{spec}) for AC and MAC calculated from galvanostatic curves at different current densities. At 500 mA g^{-1} (lowest rated value), C_{spec} values corresponded to 177.7 and 334.4 F g^{-1} for AC and MAC, respectively. Increasing the current density to 4000 mA g^{-1} (highest rated value), MAC revealed a better rate capability, retaining 80.7% of the C_{spec} value (269.9 F g^{-1}), while AC retained only 62.1% (110.4 F g^{-1}).

Coulomb and energy efficiency were also evaluated in this study (Fig. 6g and h). For both cells, coulombic efficiencies remained at 100% , while energy efficiencies were higher for cells constructed with MAC (ranging between 78.8 and 91.4% , depending on the current density) in relation to AC (ranging between 70.9 and 83.4%). The high efficiencies found for the cells testify that no degradative process occurs in great extent in the applied current and voltage ranges. This directly impacts

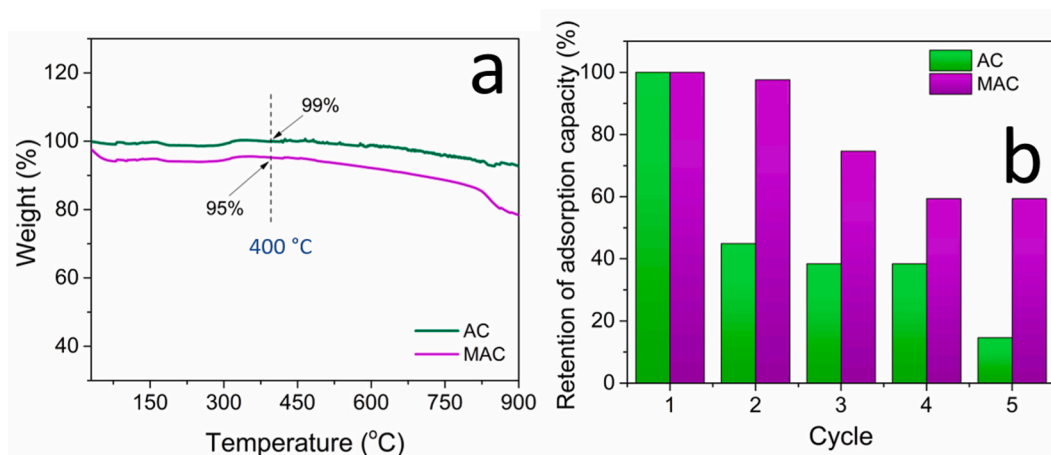


Fig. 5. (a) TGA curves for AC and MAC obtained in N_2 atmosphere. (b) Retention of MB adsorption capacity for the carbon materials as a function of thermal regeneration cycles.

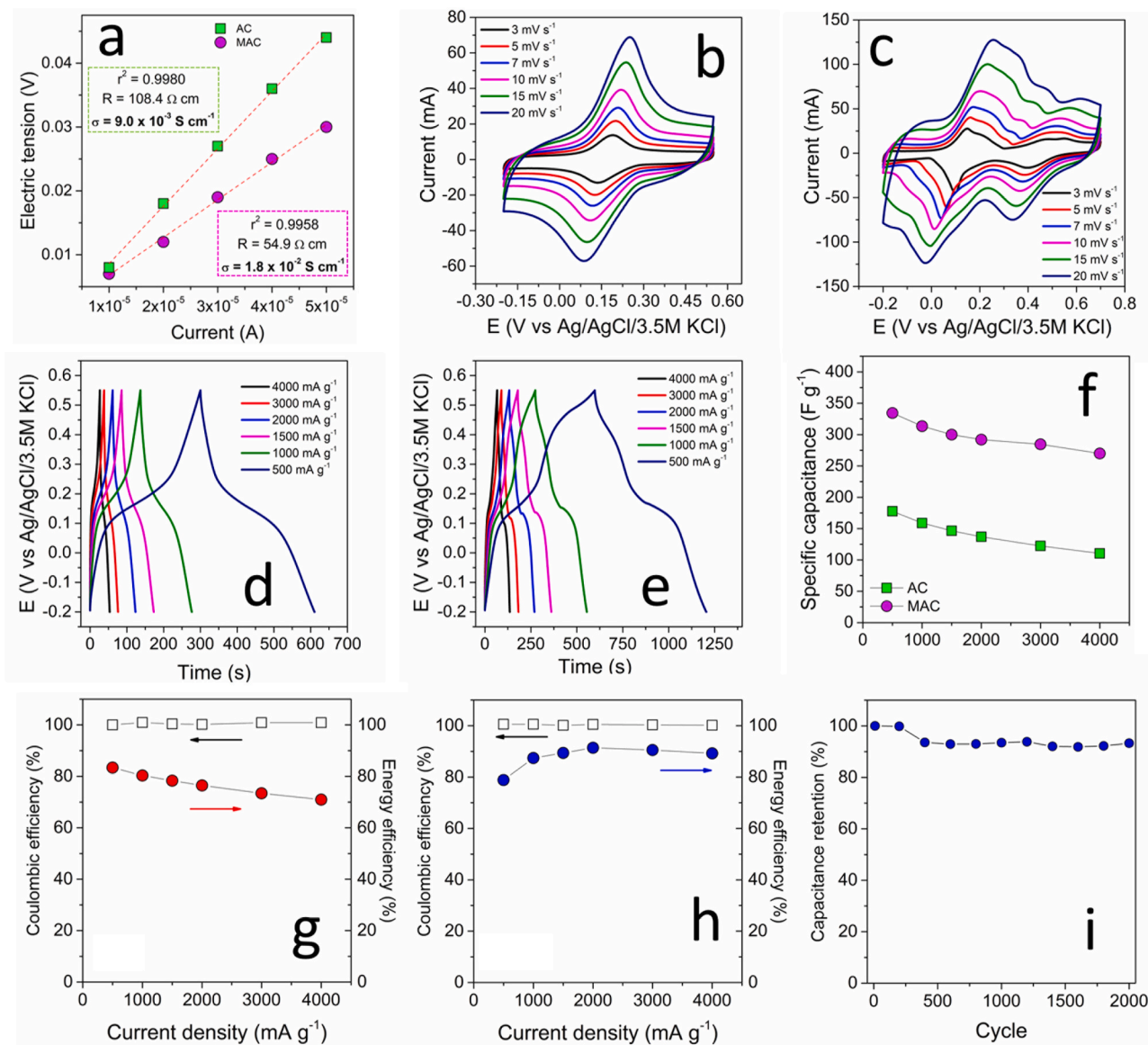


Fig. 6. (a) D.C. resistivity measurements for AC and MAC. Cyclic voltammograms obtained in a three-electrode configuration for (b) AC and (c) MAC electrodes embedded in 0.07 M MB/1M H₂SO₄. Galvanostatic measurements obtained at different current densities for (d) AC and (e) MAC electrodes. (f) Variation of specific electrode capacitance with current density. Coulombic and energy efficiencies for (g) AC and (h) MAC cells calculated at different current densities. (i) Cycling performance at 4000 mA g⁻¹ for the cell built with MAC.

cycling performance, which was evaluated for the cell constructed with MAC over 2000 cycles and showing a 93% retention of its initial capacitance (Fig. 6i). Finally, the cycling stability of the cell also reveals that the redox processes have electrochemical reversibility and there is no considerable decrease in the concentration of redox shuttles (MB²⁺/MB⁺ and Fe³⁺/Fe²⁺ couples) at the interface.

4. Conclusions

In this work, magnetic activated carbon (MAC) from waste-expanded polystyrene foam was obtained for the first time through a simple two-step conventional route. Its use as adsorbent for contaminants removal and electrode for supercapacitors demonstrates the potential applications for these non-biodegradable wastes. MAC has specific surface area and total pore volume values comparable to the best carbon adsorbents previously reported. As a consequence of these excellent textural properties, MAC showed a great capacity for adsorption of organic compounds in aqueous medium. In addition, the material can be

magnetically collected and thermally regenerated, maintaining almost 60% of its adsorption capacity after five cycles of adsorption/desorption. Applied as an electrode, MAC showed superior results in relation to non-magnetic activated carbon, with greater electrical conductivity and specific capacitance. Finally, MAC showed cycling stability with 93% retention after 2000 cycles.

CRediT authorship contribution statement

Natália C.F. Machado: Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Writing – original draft. **Larissa A.M. de Jesus:** Investigation, Methodology. **Paula S. Pinto:** Conceptualization, Data curation, Formal analysis, Supervision, Validation, Writing – original draft, Writing – review & editing. **Fabiano G.F. de Paula:** Conceptualization, Data curation, Formal analysis, Supervision, Validation, Writing – original draft, Writing – review & editing. **Márcio O. Alves:** Conceptualization, Data curation, Formal analysis, Supervision, Validation, Writing – original draft. **Kélie H.A. Mendes:**

Investigation, Methodology. **Raquel V. Mambrini**: Formal analysis, Supervision, Validation, Writing – original draft. **Daniel Barreda**: Investigation, Methodology, Data curation, Formal analysis. **Victoria Rocha**: Data curation, Formal analysis, Writing – review & editing. **Ricardo Santamaría**: Conceptualization, Supervision, Validation, Writing – review & editing. **João P.C. Trigueiro**: Conceptualization, Supervision, Funding acquisition, Validation, Writing – review & editing. **Rodrigo L. Lavall**: Conceptualization, Data curation, Funding acquisition, Supervision, Project administration, Validation, Writing – review & editing. **Paulo F.R. Ortega**: Conceptualization, Data curation, Funding acquisition, Supervision, Project administration, Validation, Writing – original draft, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.jclepro.2021.127903>.

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